Joint Trajectory Design and Power Optimization in Adversarial UAV Communications: A Game Theory Framework

Jinyue Liu, Peng Gong, Haowei Yang, Jihao Zhang, Liehuang Zhu, *Senior Member, IEEE*, and Xiang Gao

***Abstract*—To address the challenges of dynamic jamming and resource optimization in uncrewed aerial vehicle (UAV) confrontation communication networks, this article proposes a hierarchical game framework for joint trajectory design and power allocation. These elements are inherently coupled: jamming degrades channel quality, prompting UAVs to adapt both their trajectories to improve spatial positioning and reduce jamming exposure, and their power allocation to maintain communication effectiveness under interference. The framework captures this interplay through a differential game for trajectory optimization and a Stackelberg game for power allocation, with the Nash equilibrium (NE) rigorously defined and its existence proved. To overcome the complexity of solving the Hamilton–Jacobi–Bellman (HJB) equation, an adaptive dynamic programming (ADP) algorithm is employed to approximate the value functions for both trajectory and power optimization. Simulations validate the effectiveness of the proposed framework, demonstrating that it accurately captures the strategic interactions between UAVs and provides robust decision support for resource optimization in dynamic adversarial environments.**

***Index Terms*—Adaptive dynamic programming (ADP), anti-jamming, differential game, Stackelberg game, uncrewed aerial vehicle (UAV).**

I. Introduction

UNCREWED aerial vehicles (UAVs) have become indispensable in modern communication networks, providing unparalleled flexibility, mobility, and coverage. Their capability to function as aerial base stations or relay nodes has significantly advanced wireless communication [1], [2], [3], [4], [5], [6], especially in scenarios demanding rapid deployment or extended coverage, such as disaster relief [7], [8] and military operations [9], [10].

The inherent openness of aerial communications, coupled with the constrained resources and complex signal propagation environments, renders UAV networks highly susceptible to jamming attacks [11]. Adversaries can exploit this vulnerability by transmitting high-power jamming signals, effectively disrupting communication between UAVs and ground stations or among UAVs themselves. Such disruptions can lead to data transmission failures, navigation signal loss, and, in severe cases, the complete loss of UAV control or catastrophic failures. In recent years, extensive research has been conducted on anti-jamming techniques for UAV communications, covering various aspects such as multi-UAV cooperation [12], [13], intelligent reflecting surfaces (IRSs) selection [14], [15], [16], [17], UAV trajectory planning [18], [19], [20], [21], as well as UAV swarm architecture design [22], resource allocation [23], [24], UAV scheduling [25], [26], and interference suppression techniques [27], [28] and cryptographic algorithm [29], [30], [31], [32].

While previous studies have mainly focused on static interference patterns [14], [18], [22], [23], [25], [33], [34], recent advancements in adaptive jamming technologies enable attackers to dynamically adjust their strategies based on UAV communication characteristics [35], [36], [37]. These adaptive jammers can intelligently change parameters such as frequency, power, and modulation schemes in real time, making them significantly more effective than traditional static jammers while simultaneously evading conventional detection methods. As a result, UAV communication systems face increased risks of signal disruption, degraded data integrity, and compromised mission-critical operations. Existing anti-jamming mechanisms, which are primarily designed to counter static or predictable interference, often lack the responsiveness, flexibility, and real-time decision-making capability required to mitigate such sophisticated attacks. For instance, methods that rely on threshold-based detection may fail to identify subtle variations in signal characteristics introduced by adaptive jammers. Consequently, these conventional approaches are insufficient to maintain reliable communications under dynamic and intelligent jamming scenarios. These challenges underscore the urgent need for intelligent and adaptive anti-jamming strategies that can dynamically respond to evolving threats, ensuring reliable and secure UAV communications in contested electromagnetic environments.

To address these challenges, this article proposes a hierarchical game-theoretic framework for joint trajectory design and power allocation in UAV confrontation networks. In adversarial UAV communications, jamming, resource allocation,

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and trajectory design are inherently interrelated: the jammer's interference degrades channel quality, which directly influences power control decisions, while the UAV's trajectory determines spatial exposure to jamming. Effective anti-jamming performance, therefore, requires joint optimization of movement and power strategies within a unified framework. We investigate the jamming and anti-jamming dynamics in a UAV tactical communication network, where a UAV base station serves distributed ground units, while an intelligent cognitive UAV jammer actively disrupts legitimate communications between the UAV base station and ground units. The interaction between the UAV base station and the adversarial jammer unfolds as a pursuit-evasion game in a time-varying environment. In this adversarial setting, the UAV base station must execute dynamic evasive maneuvers to counteract intelligent jamming attacks while maintaining connectivity with distributed ground units. To achieve this, it employs an integrated approach to trajectory planning and power control, striving to minimize both jamming exposure and energy consumption. Conversely, the UAV jammer adaptively optimizes its pursuit trajectory and power allocation to maximize communication disruption within its energy constraints. The principal innovation of this study resides in the formulation of a hierarchical game-theoretic framework that systematically decouples the original coupled optimization problem into two interactive subproblems. A hierarchical game is used because UAV networks involve multilevel interactions: decisions at the power allocation level influence trajectory planning, and vice versa. This layered structure explicitly captures the bidirectional coupling between resource allocation and trajectory optimization, ensuring that both levels respond adaptively to the evolving jamming environment. Trajectory dynamics are modeled using a differential game, which captures real-time pursuit-evasion interactions subject to aerodynamic constraints. Power allocation is analyzed through a Stackelberg game, where the UAV jammer acts as the leader that strategically anticipates and optimally responds to the UAV-BS's anti-jamming strategies. Compared to existing research that typically considers either static strategies or single-level interactions, our hierarchical framework integrates multilevel decision-making with dynamic adaptation to intelligent jamming, providing a more effective and realistic anti-jamming solution. To solve the equilibrium of the proposed game, this article employs an adaptive dynamic programming (ADP) approach to optimize the trajectory and power design of the UAV base station and UAV jammer. This framework not only enhances the resilience of UAV-based communication systems against adversarial interference but also provides valuable theoretical insights for future UAV confrontation network designs. This integrated design reveals the intrinsic coupling between trajectory planning, power control, and jamming mitigation, providing a coherent foundation for adaptive decision-making in adversarial UAV networks. The main contributions of this article are as follows.

1) A hierarchical game-theoretic framework is designed to decompose the joint trajectory and power optimization problem in UAV networks into two subproblems: a differential game for trajectory planning and a Stackelberg game for power allocation. Leveraging a dynamic strategy transfer mechanism, the framework reduces the complexity of the multidimensional decision space.
2) A differential game is employed to characterize the real-time pursuit-evasion dynamics between the UAV base station and the UAV jammer at the trajectory optimization layer. By integrating kinematic equations with energy constraint equations, a multidomain modeling framework is established that captures both the motion dynamics and energy consumption characteristics. At the power allocation layer, leveraging the UAV jammer's proactive strategy prediction along with the UAV base station's reciprocal feedback mechanism, traditional static channel competition is transformed into an adaptive equilibrium process based on bidirectional strategy responses, thereby overcoming the inherent limitations of rigid jamming modes in existing research.
3) To address the complexity of Hamilton–Jacobi–Bellman (HJB) equation solutions, we adopt a neural network-based ADP algorithm. By using dual-channel critic networks to independently approximate the value functions of trajectory and power games, the method ensures the existence of a Nash equilibrium (NE) while achieving rapid convergence in strategy space under complex adversarial environments. This framework provides a decision-making solution for dynamic gaming scenarios.

The rest of this article is organized as follows. Section II discusses the related works. Section III is about the system model and problem formulation. Analysis for the NE of the game is given in Section IV. Section V presents the proposed ADP-based method. Numerical simulations are provided in Section VI. Finally, Section VII gives a conclusion of this work.

II. Related Works

Recent years have seen significant research on addressing jamming threats in UAV communication networks. Studies have examined various aspects, including IRSs, multi-UAV collaboration, and resource allocation strategies, particularly in UAV trajectory planning, channel assignment, power allocation, and joint optimization approaches.

Duo et al. [18] study the anti-jamming 3-D trajectory design for UAVs in wireless sensor networks (WSNs). The UAV, deployed to collect data from multiple ground sensors (GSs) while counteracting malicious ground-based jammers, maximizes the minimum average expected data collection rate of GSs within a constrained flight duration. This optimization is achieved through the joint design of GS transmission scheduling and UAV trajectory planning. Wu et al. [22] investigate communication challenges in large-scale UAV swarms under jamming environments, with the objective of maximizing the total data transmission rate of cluster-head UAVs. A coordinated optimization framework integrating joint trajectory design and cluster formation is proposed. Wu et al. [23] address the performance optimization of UAV relay communication systems under malicious interference by proposing a joint optimization of UAV trajectory and transmission power to

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enhance end-to-end communication throughput. Mathematical models incorporating UAV flight constraints and transmission power limitations are constructed, achieving a significant improvement in the destination node's throughput under limited flight time. Deng et al. [25] investigate UAV communication system optimization under multiple interference sources and imperfect channel state information (CSI). By jointly optimizing UAV scheduling strategies, power allocation schemes, and 3-D trajectory planning, the proposed method maximizes the average data rate for communications with all GSs while counteracting interference. Wang et al. [38] focus on the design of an energy-constrained UAV-assisted secure communication system. While acting as a mobile relay to support communication, the UAV also employs artificial noise (AN) to interfere with eavesdroppers. To enhance security and reliability, a joint optimization scheme is proposed to optimize UAV positioning, AN transmission power, and the power splitting and time switching ratios. To tackle the nonconvex optimization problem, the authors decompose it into three subproblems and solve them separately. Wen et al. [39] propose a multiagent reinforcement learning approach to optimize UAV trajectories and power for secure communication, focusing on enhancing physical layer security by preventing eavesdropping with cooperative jamming.

However, most existing studies fall short in modeling the dynamic characteristics of jammers, often relying on fixed-location assumptions or predefined interference patterns. With advancements in antenna technology and cognitive electronic warfare systems, modern jammers now exhibit dynamic behavior, including physical mobility and adaptive strategy adjustments. Simplified models struggle to accurately capture real-world adversarial scenarios, and traditional anti-jamming methods become significantly less effective when confronted with jammers capable of cognitive decision-making.

To address the aforementioned challenges, game theory, as a powerful tool for analyzing interactions among decision-makers, offers novel theoretical perspectives and solutions for communication confrontation problems. The core idea is to model the communication system and the jammer as interdependent decision-making entities. By analyzing their strategy selection and corresponding payoff functions, optimal countermeasure strategies can be derived to enhance system performance and robustness.

To realize the collaboration and competition among UAVs in a distributed manner, Yin et al. [40] demonstrate that the formulated joint channel-power optimization problem constitutes an exact potential game and proposes a collaborative multiagent layered Q-learning (MALQL)-based anti-jamming communication algorithm to reduce the high dimensionality of the action space and accelerate the algorithm convergence speed. Xiao et al. [41] present a systematic investigation of relay strategies in UAV-assisted vehicular ad hoc networks (VANETs). The study employs a stochastic game-theoretic framework to characterize the strategic interaction between UAV relays and jammers, ultimately developing robust relay countermeasures against malicious interference in vehicular networks. Fan et al. [42] address the problem of hybrid attacks in UAV communication networks by formulating an adversarial game model with partial observation characteristics. The UAV trajectory is optimized to achieve an equilibrium between the UAV and the jammer. Han et al. [43] investigate the problem of intelligent jamming in the satellite-enabled army IoT (SaIoT) and propose a hierarchical anti-jamming Stackelberg game (HASG) model to reduce energy consumption in jamming environments. In this model, the jammer acts as the leader, selecting the optimal jamming location, while the SaIoT devices, as followers, counteract the interference by forming communication subnets. Furthermore, an anti-jamming coalition formation game (CFG) and a reinforcement learning algorithm are introduced to enable distributed optimization and facilitate stable coalition formation.

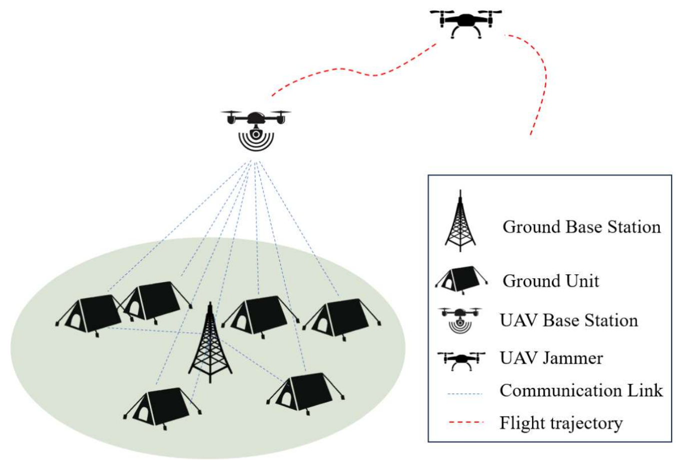

Fig. 1. UAV confrontation communication scenario.

This article proposes a hierarchical game framework that decomposes UAV adversarial decision-making into two optimization layers. At the trajectory planning layer, a differential game is employed to model real-time pursuit-evasion dynamics, enabling maneuvering evasion responses through continuous-time strategies. At the power allocation layer, a leader–follower decision paradigm is formulated based on the Stackelberg game, where the UAV jammer (leader) and the UAV base station (follower) engage in strategic interactions. This framework not only reduces the complexity of the joint strategy space but also enhances computational efficiency, ensuring feasibility for resource-constrained airborne computing platforms.

III. System Model and Problem Formulation

We consider an UAV confrontation communication scenario where one UAV-enabled base station communicates with N ground units using frequency division multiple access, and one malicious UAV-enabled jammer attempts to intelligently jam the link from the UAV-enabled base station to ground units with the tracking jamming method, as shown in Fig. 1. For simplicity, we use the base station and the jammer to refer to the UAV-enabled base station and the UAV-enabled jammer in the following parts. The set of the ground mobile units is denoted as $\mathbf{N}$. Within the allowable range, the UAV-enabled jammer adjusts its trajectory and jamming power to form an effective jam, while the UAV-BS can adjust its trajectory

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and sending power to reduce the influence caused by the UAV-jammer. The base station and the jammer move in the space, formulating the "pursuit-evasion" interaction. The game is described as

$$\text{Game} = \{\text{BS}, J, A_{\text{BS}}, A_J, u_{\text{BS}}, u_J\} \tag{1}$$

where BS denotes the base station; J denotes the jammer; A_{BS} is the strategy space of the base station; and A_J is the strategy space of the jammer. u_{BS} and u_j denote the utility of the base station and the jammer, respectively.

A. Communication Model

We assume that the base station is equipped with directional antennas. The antenna gain of the base station can be denoted as follows:

$$G = \begin{cases} G^{3\,\text{dB}}, & -\frac{\theta}{2} \le \varphi \le \frac{\theta}{2} \\ [2ex]\, G(\varphi), & \text{otherwise} \end{cases} \tag{2}$$

where θ is the beamwidth, φ is the sector angle, $G^{3\,\text{dB}}$ is the gain of main lobe, and the $G(\varphi)$ is the gain of sidelobe.

Without loss of generality, we consider a scenario where there is both a line of sight (LOS) and a non-line-of-sight (NLOS) link between the base station and ground units. The path loss can be denoted as follows:

$$L_{\text{LOS}}(t) = \alpha \cdot \left(d_{\text{BS},n}(t)\right)^{-\beta} \tag{3}$$

where α is the path loss coefficient at the reference distance of 1 m, $d_{\text{BS}n}(t)$ is the distance between the base station and the ground unit n at the time instant t, and $\beta \ge 2$ is the path loss exponent. When the link belongs to the LOS link, $\beta = 2$.

We denote the probability of the LOS link as p_{LOS}, then the probability of the NLOS link is $p_{\text{NLOS}} = 1 - p_{\text{LOS}}$. Thus, the average path loss of the link between the base station and the nth ground unit can be denoted as follows:

$$\begin{aligned} L_{\text{BS}}^n &= p_{\text{LOS}} L_{\text{LOS}} + p_{\text{NLOS}} L_{\text{NLOS}} \\ &= p_{\text{LOS}}\alpha \cdot \left(d_{\text{BS},n}(t)\right)^{-2} + (1 - p_{\text{LOS}})\,\alpha \cdot \left(d_{\text{BS},n}(t)\right)^{-\beta}. \end{aligned} \tag{4}$$

We define the jamming power of the jammer as $P_{\text{jamm}}(t)$ and the channel gain from the jammer to the nth ground unit as $g_{\text{jamm}}^n(t) = GL_{\text{jamm}}^n$, where L_{jamm}^n has similar definition to L_{BS}^n. And we assume that the UAV-jammer can only jam one channel at the same time. Then, the interference caused by the jammer is denoted as $I_{\text{jamm}}^n = \delta(l_n, j) P_{\text{jamm}}(t) g_{\text{jamm}}^n(t)$, in which $\delta(x_1, x_2) = \begin{cases} 1, x_1 = x_2 \\ 0, x_1 \ne x_2 \end{cases}$ means only when the channel l_n chosen by the base station to communicate with the ground unit n is same with the jamming channel j, the interference is considered.

We denote that the sending power of the base station is $P_{\text{send}}(t)$, the bandwidth is B, and the noise power is N. Then, the achievable rate between the base station and the nth ground unit is

$$R_{\text{BS}}^n(t) = B \log_2 \left(1 + \frac{P_{\text{send}}(t)\, G L_{\text{BS}}^n}{I_{\text{jamm}}^n(t) + N}\right). \tag{5}$$

We define the success rate of the jamming as $R_J^n(t) = R_{\text{BS}_0}^n(t) - R_{\text{BS}}^{\text{n}}(t)$, where $R_{\text{BS}_0}^n(t)$ is the achievable rate between the base station and the nth unit when there is no jammer, i.e., $I_{\text{jamm}}^n(t) = 0$.

B. Motion Model and Energy Model

The coordinates of N ground units at any time instant t are denoted by $\mathbf{l}_{\text{n}}(t) = [x_n(t), y_n(t)], n \in \mathbf{N}$, which is a 2-D coordinates. And the flight heights of the base station and the UAV-jammer are the same and fixed, which are denoted by H. We denote $\mathbf{l}_{\text{BS}}(t) = [x_{\text{BS}}(t), y_{\text{BS}}(t)]$, $\mathbf{v}_{\text{BS}}(t) = \left[v_{\text{BS}}^x(t), v_{\text{BS}}^y(t)\right]$ and $\mathbf{a}_{\text{BS}}(t) = \left[a_{\text{BS}}^x(t), a_{\text{BS}}^y(t)\right]$ as the 2-D coordinates, velocity and acceleration of the base station at time instant t, respectively. Similarly, we label $\mathbf{l}_J(t) = [x_J(t), y_J(t)]$, $\mathbf{v}_J(t) = \left[v_J^x(t), v_J^y(t)\right]$ and $\mathbf{a}_J(t) = [a_J^x(t), a_J^y]$ as the 2-D coordinates, velocity and acceleration of the jammer at time instant t, respectively. Therefore, the motion equation of the base station can be expressed as follows:

$$\begin{cases} \dot{\mathbf{l}}_{\text{BS}}(t) = \mathbf{v}_{\text{BS}}(t) \\ \dot{\mathbf{v}}_{\text{BS}}(t) = \mathbf{a}_{\text{BS}}(t). \end{cases} \tag{6}$$

And the initial conditions can be denoted as follows:

$$\begin{cases} \mathbf{l}_{\text{BS}}(0) = \left[x_{\text{BS}_0}, y_{\text{BS}_0}\right] \\ \mathbf{v}_{\text{BS}}(0) = \left[v_{\text{BS}_0}^x, v_{\text{BS}_0}^y\right]. \end{cases} \tag{7}$$

This means that the base station has a fixed initial location and velocity, while allowing for free final states.

Similarly, the motion equation of the UAV-jammer can be expressed as follows:

$$\begin{cases} \dot{\mathbf{l}}_{\mathbf{J}}(t) = \mathbf{v}_{\mathbf{J}}(t) \\ \dot{\mathbf{v}}_{\mathbf{J}}(t) = \mathbf{a}_{\mathbf{J}}(t) \end{cases} \tag{8}$$

$$\begin{cases} \mathbf{l}_{\mathbf{J}}(0) = \left[x_{J_0}, y_{J_0}\right] \\ \mathbf{v}_{\mathbf{J}}(0) = \left[v_{J_0}^x, v_{J_0}^y\right] \end{cases}. \tag{9}$$

Suppose that the mass of both the base station and the UAV-jammer is m, the force provide by the engine is $\mathbf{F}$, and the aerodynamic resistance experienced by a UAV during flight is $\mathbf{R}_*(* \in \{\text{BS}, J\})$, where $\mathbf{R}_* = -(1/2)\rho C_d S \|\mathbf{v}_*(t)\|^2$ is the air density, C_d is the resistance coefficient and S is the UAV's effective cross-sectional area. According to Newton's second law, we have $\mathbf{F}_*(t) + \mathbf{R}_*(t) = m\mathbf{a}_*(t)$. Therefore, during the time interval $(t, t + dt)$, the motion power of UAV can be approximately expressed as [44]

$$\begin{aligned} &\mathbf{F}_\star(t)\, \mathrm{d}\mathbf{s}_\star(t)^T \\ &= \left(m\mathbf{a}_\star(t) + \frac{1}{2}\rho C_d S \|\mathbf{v}_\star(t)\|^2\right)\left(\mathbf{v}_\star(t)^T\ dt + \frac{\mathbf{a}_\star(t)^2}{2}(dt)^2\right) \\ &= m\mathbf{a}_\star(t)\,\mathbf{v}_\star(t)^T\ dt + \frac{1}{2}\rho C_d S \|\mathbf{v}_\star(t)\|^2 \mathbf{v}_\star(t)^T\ dt \\ &+ \frac{1}{2}m\|\mathbf{a}_\star(t)\|^2 (dt)^2 + \frac{1}{4}\rho C_d S \|\mathbf{v}_\star(t)\|^2 \mathbf{a}_\star(t)^T (dt)^2 \\ &= m\mathbf{a}_\star(t)\,\mathbf{v}_\star(t)^T\ dt + C\|\mathbf{v}_\star(t)\|^2 \mathbf{v}_\star(t)^T\ dt \\ &+ \frac{1}{2}m\|\mathbf{a}_*(t)\|^2 (dt)^2 + \frac{1}{2}C\|\mathbf{v}_*(t)\|^2 \mathbf{a}_*(t)^T (dt)^2. \end{aligned} \tag{10}$$

Based on (10), we define the motion power of the base station $P_{\text{move,BS}}(t)$ as

$$P_{\text{move,BS}}(t) = m\mathbf{a}_{\text{BS}}(t)\,\mathbf{v}_{\text{BS}}(t)^T + C\|\mathbf{v}_{\text{BS}}(t)\|^2 \mathbf{v}_{\text{BS}}(t)^T$$

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$$+\ \frac{1}{2}m\|\mathbf{a}_{\mathrm{BS}}(t)\|^2+\frac{1}{2}C\|\mathbf{v}_{\mathrm{BS}}(t)\|^2\mathbf{a}_{\mathrm{BS}}(t)^T. \quad (11)$$

Then, we can get the energy equation of the base station, which is denoted as

$$\dot{e_{\mathrm{BS}}}(t)=P_{\mathrm{send}}(t)+P_{\mathrm{move,BS}}(t) \quad (12)$$

and the initial condition of the energy model is given as

$$e_{\mathrm{BS}}(0)=e_{\mathrm{BS}_0}. \quad (13)$$

Similarly, we have the energy model of the UAV-jammer as

$$\begin{cases} \dot{e_J}(t)=P_{\mathrm{jamm}}(t)+P_{\mathrm{move},J}(t)\\ e_J(0)=e_{J_0}. \end{cases} \quad (14)$$

C. Problem Formulation

Different from the terrestrial base station, the onboard energy of the base station is limited, so the objective of the base station is to find the optimal trajectory design and power allocation strategy to maximize the achievable rate under the jammer situation and min the power consumption simultaneously. The power consumption includes the power consumption for data transmission and motion. Based on the communication model and the motion model we defined above, we can denote the utility as follows:

$$\begin{aligned} &u_{\mathrm{BS}}\left(\mathbf{a}_{\mathrm{BS}}(t),P_{\mathrm{send}}(t),\mathbf{a}_{\mathbf{J}}(\mathbf{t}),P_{\mathrm{jamm}}(t)\right)\\ &=\int_{t=0}^{T}\sum_{n=1}^{N}R_{\mathrm{BS}}^{n}(t)\,dt-\eta_1\int_{t=0}^{T}P_{\mathrm{send}}(t)\,dt\\ &\quad-\ \eta_2\int\mathbf{F}_{\mathrm{BS}}(t)\,\mathrm{d}s_{\mathrm{BS}}(t)^{\mathrm{T}}\\ &=\int_{t=0}^{T}\left(\sum_{n=1}^{N}R_{\mathrm{BS}}^{n}(t)-\eta_1P_{\mathrm{send}}(t)-\eta_2P_{\mathrm{move,BS}}(t)\right)dt\\ &=\int_{t=0}^{T}g_{\mathrm{BS}}(t)\,dt \end{aligned} \quad (15)$$

where the $T>0$ denotes the terminal time instant of the game. η_1 and η_2, respectively, denote the weight coefficients for the data transmission power consumption and motion power consumption.

Then, the joint optimization problem of the base station can be expressed as

$$\begin{aligned} \max_{a_{\mathrm{BS}},P_{\mathrm{send}}}\ & u_{\mathrm{BS}}\left(\mathbf{a}_{\mathrm{BS}}(t),P_{\mathrm{send}}(t),\mathbf{a}_{\mathbf{J}}(t),P_{\mathrm{jamm}}(t)\right)\\ \text{s.t.}\ & P_{\mathrm{BS,min}}\le P_{\mathrm{send}}(t)\le P_{\mathrm{BS,max}}\quad\forall t\\ & \mathbf{a}_{\mathrm{BS,min}}\le\mathbf{a}_{\mathrm{BS}}(t)\le\mathbf{a}_{\mathrm{BS,max}}\quad\forall t. \end{aligned} \quad (16)$$

As for the jammer, its purpose is to maximize the data transmission jammer success rate and min the power consumption simultaneously by finding the optimal trajectory design and power allocation strategy, where the power consumption is decided by the jammer power and motion power. Similar to a base station, we can denote the utility of the jammer as follows:

$$\begin{aligned} &u_{\mathrm{J}}\left(\mathbf{a}_{\mathrm{BS}}(t),P_{\mathrm{send}}(t),\mathbf{a}_{\mathbf{J}}(t),P_{\mathrm{jamm}}(t)\right)\\ &=\int_{t=0}^{T}\sum_{n=1}^{N}R_{J}^{n}(t)\,dt-\mu_1\int_{t=0}^{T}P_{\mathrm{jamm}}(t)\,dt\\ &\quad-\ \mu_2\int\mathbf{F}_{\mathbf{J}}(t)\,ds_J(t)^{\mathrm{T}}\\ &=\int_{t=0}^{T}\left(\sum_{n=1}^{N}\left(R_{\mathrm{BS}_0}^{n}(t)-R_{\mathrm{BS}}^{n}(t)\right)-\mu_1P_{\mathrm{jamm}}(t)\right.\\ &\qquad\left.-\mu_2P_{\mathrm{move},J}(t)\right)dt\\ &=\int_{t=0}^{T}g_J(t)\,dt \end{aligned} \quad (17)$$

where μ_1 and μ_2, respectively, denote the weight coefficients for the jammer consumption and motion power consumption.

Then, the joint optimization problem of the jammer can be expressed as

$$\begin{aligned} \max_{a_J,P_{\mathrm{jamm}}}\ & u_J\left(\mathbf{a}_{\mathrm{BS}}(t),P_{\mathrm{send}}(t),\mathbf{a}_{\mathbf{J}}(t),P_{\mathrm{jamm}}(t)\right)\\ \text{s.t.}\ & P_{\mathrm{jamm,min}}\le P_{\mathrm{jamm}}(t)\le P_{\mathrm{jamm,max}}\quad\forall t\\ & \mathbf{a}_{J,\mathrm{min}}\le\mathbf{a}_J(t)\le\mathbf{a}_{J,\mathrm{max}}\quad\forall t. \end{aligned} \quad (18)$$

IV. Equilibrium Analysis of Hierarchical Differential Game

To reduce the complexity of solving the joint optimal problem, we propose a hierarchical game model, dividing the joint game into two subgames. As shown in Fig. 2, the framework consists of two parts: one is the upper layer trajectory subgame, and the other is the lower layer power subgame. The upper layer subgame is responsible for making an acceleration decision, and then it passes its decision to the lower layer subgame. The lower layer subgame decides the power optimal strategy according to the players' position, which from the upper layer subgame, and gives the results to the upper layer subgame as feedback. Then, this process is continuously looped until the equilibrium of the joint game is found.

A. Upper Layer: Differential Game of Trajectory

We assume that the sending power of the base station and the jamming power of the jammer are fixed in the upper layer subgame. Under this assumption, we denote the trajectory design subgame as $G_u=\{\mathrm{BS},J,\mathbf{a}_{\mathrm{BS}},\mathbf{a}_{\mathbf{J}},u_{\mathrm{BS}},u_J\}$, where BS and J are the base station and jammer, respectively. $\mathbf{a}_{\mathrm{BS}}$ denotes the acceleration strategy of the base station and the $\mathbf{a}_{\mathbf{J}}$ denotes the acceleration strategy of the jammer. u_{BS} is shown as in (15) and u_J is shown as in (17). Note that and are only related to the acceleration strategy and do not consider the power strategy effects. Next, we give a definition and solution of the NE for this game.

Definition 1 (NE): For the game $G_u=\{B,J,\mathbf{a}_{\mathrm{BS}},\mathbf{a}_J,u_{\mathrm{BS}},u_J\}$, strategy combination $\{\mathbf{a}_{\mathrm{BS}}^{\star},\mathbf{a}_J^{\star}\}$ is the NE if and only if

$$\begin{aligned} u_{\mathrm{BS}}\left(\mathbf{a}_{\mathrm{BS}}^{*},\mathbf{a}_J^{*}\right)&\ge u_{\mathrm{BS}}\left(\mathbf{a}_{\mathrm{BS}},\mathbf{a}_J^{*}\right)\\ u_J\left(\mathbf{a}_{\mathrm{BS}}^{*},\mathbf{a}_J^{*}\right)&\ge u_J\left(\mathbf{a}_{\mathrm{BS}}^{*},\mathbf{a}_J\right) \end{aligned} \quad (19)$$

where $\mathbf{a}_{\mathrm{BS}}^{\star}$ is the optimal acceleration strategy of the base station and $\mathbf{a}_J^{\star}$ is the optimal acceleration strategy of the jammer.

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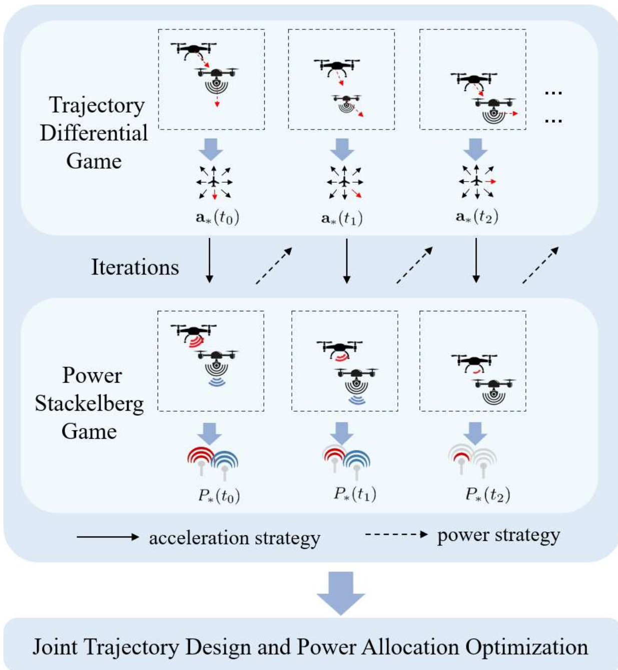

Fig. 2. Diagram of the joint optimization method for trajectory design and power allocation.

Equation (19) indicates that, at the NE, either the base station or the jammer cannot get a higher utility by unilaterally changing its strategy.

Lemma 1 (Existence of Optimal Solution): For any fixed opponent strategy, each player's optimization problem admits at least one optimal solution.

Proof: We prove the existence of the base station; the jammer's case follows by symmetry. We verify the conditions for existence using the Weierstrass theorem:

1) *Compact Feasible Set:* The control constraints form a compact set

$$A_{\mathrm{BS}} = \left\{ \begin{array}{l} (\mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}): \\ \quad \mathbf{a}_{\mathrm{BS,min}} \leq \mathbf{a}_{\mathrm{BS}} \leq \mathbf{a}_{\mathrm{BS,max}}, \\ \quad P_{\mathrm{BS,min}} \leq P_{\mathrm{send}} \leq P_{\mathrm{BS,max}} \end{array} \right\}. \tag{20}$$

2) *Bounded Trajectories:* The state trajectories remain bounded on $[0, T]$ due to

$$\|\mathbf{v}_{\mathrm{BS}}(t)\| \leq \|\mathbf{v}_{\mathrm{BS}_0}\| + \mathbf{a}_{\mathrm{BS,max}} T \tag{21}$$

$$\|\mathbf{l}_{\mathrm{BS}}(t)\| \leq \|\mathbf{l}_{\mathrm{BS}_0}\| + \left(\|\mathbf{v}_{\mathrm{BS}_0}\| + \mathbf{a}_{\mathrm{BS,max}} T\right) \tag{22}$$

$$0 \leq e_{\mathrm{BS}}(t) \leq e_{\mathrm{BS}_0}. \tag{23}$$

The energy constraint ensures that the set of admissible control trajectories satisfying the dynamics is compact.

3) *Continuity of Objective Function:* From (15), the utility function u_{BS} is continuous in $(a_{\mathrm{BS}}, P_{\mathrm{send}})$ because as follows.

a) Data rate $R^n_{\mathrm{BS}}(t)$ is continuous in positions and powers.

b) Motion power $P_{\mathrm{move,BS}}$ from (11) is a polynomial in $v_{\mathrm{BS}}, a_{\mathrm{BS}}$.

4) *Lipschitz Continuity of Dynamics:* The system dynamics are Lipschitz continuous with constant $L = 1 + \|\mathbf{a}_{\mathrm{BS,max}}\|$. Specifically, for state vectors

$$\mathbf{x}_{\mathrm{BS}}(t) = \begin{bmatrix} \mathbf{l}_{\mathrm{BS}}(t) \\ \mathbf{v}_{\mathrm{BS}}(t) \end{bmatrix} \tag{24}$$

$$\|\dot{\mathbf{x}}_{\mathrm{BS}}(t) - \dot{\mathbf{x}}'_{\mathrm{BS}}(t)\| \leq L\|\mathbf{x}_{\mathrm{BS}}(t) - \mathbf{x}'_{\mathrm{BS}}(t)\|. \tag{25}$$

The dynamics are Lipschitz continuous because the position and velocity are continuous, and their derivatives (acceleration) are also bounded. Specifically, the bounded acceleration ensures that the rate of change of velocity is limited, which guarantees the Lipschitz continuity of the system.

By the Weierstrass extreme value theorem, a continuous function on a compact set attains its maximum. Therefore, an optimal solution exists for the base station. The proof for the jammer follows the same argument with the corresponding control set and constraints. □

Lemma 2: If the acceleration strategy $\mathbf{a}^{\star}_{\mathrm{BS}}$ and $\mathbf{a}^{\star}_{J}$ constitute a NE, then there exists a continuously differentiable function $J_{\mathrm{BS}}(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, e_{\mathrm{BS}}, t): R^2 \times R^2 \times [0, E_{\mathrm{send,max}}] \times [0, T] \rightarrow R$ that satisfy the following HJB equation:

$$\begin{aligned} -\nabla J_{\mathrm{BS}}(t) = \max_{\mathbf{a}_{\mathrm{BS}}} \big[& g_{\mathrm{BS}}(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, \mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}, t) \\ & + \nabla J_{\mathrm{BS}}(\mathbf{l}_{\mathrm{BS}}) \cdot \mathbf{v}_{\mathrm{BS}} + \nabla J_{\mathrm{BS}}(\mathbf{v}_{\mathrm{BS}}) \cdot \mathbf{a}_{\mathrm{BS}} \\ & + \nabla J_{\mathrm{BS}}(e_{\mathrm{BS}}) \cdot \left(P_{\mathrm{send}} + P_{\mathrm{move,BS}}\right) \big] \end{aligned} \tag{26}$$

and there exists a continuously differentiable function $J_J(\mathbf{l}_J, \mathbf{v}_J, e_J, t): R^2 \times R^2 \times [0, E_{\mathrm{jamm,max}}] \times [0, T] \rightarrow R$ which satisfies the equation as follows:

$$\begin{aligned} -\nabla J_J(t) = \max_{\mathbf{a}_J} \big[& g_J(\mathbf{l}_J, \mathbf{v}_J, \mathbf{a}_J, P_{\mathrm{jamm}}, t) \\ & + \nabla J_J(\mathbf{l}_J) \cdot \mathbf{v}_J + \nabla J_J(\mathbf{v}_J) \cdot \mathbf{a}_J \\ & + \nabla J_J(e_J) \cdot (P_{\mathrm{jamm}} + P_{\mathrm{move},J}) \big]. \end{aligned} \tag{27}$$

Proof: Please refer to Appendix A for detailed proof.

We define the Hamiltonian function of the base station as follows:

$$\begin{aligned} H_{\mathrm{BS}} \triangleq\; & g_{\mathrm{BS}}(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, \mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}, t) \\ & + \nabla J_{\mathrm{BS}}(\mathbf{l}_{\mathrm{BS}}) \cdot \mathbf{v}_{\mathrm{BS}} + \nabla J_{\mathrm{BS}}(\mathbf{v}_{\mathrm{BS}}) \cdot \mathbf{a}_{\mathrm{BS}} \\ & + \nabla J_{\mathrm{BS}}(e_{\mathrm{BS}}) \cdot \left(P_{\mathrm{send}} + P_{\mathrm{move,BS}}\right). \end{aligned} \tag{28}$$

Furthermore, according to the necessary condition of the optimal principle, the optimal acceleration strategy can be obtained by setting $(\partial H_{\mathrm{BS}} / \partial \mathbf{a}_{\mathrm{BS}}) = 0$ and it is given by

$$\mathbf{a}^{*}_{\mathrm{BS}} = -\mathbf{v}_{\mathrm{BS}} - C\frac{\|\mathbf{v}_{\mathrm{BS}}\|^2}{m} + \frac{\nabla J_{\mathrm{BS}}(\mathbf{v}_{\mathrm{BS}})}{m(\eta_2 - \nabla J_{\mathrm{BS}}(e_{\mathrm{BS}}))}. \tag{29}$$

The optimizer $\mathbf{a}^{*}_{\mathrm{BS}}$ is well defined, and its local uniqueness follows from the first-order condition together with the second-order sufficient condition; see Lemma 3 for the precise diagonal dominance condition.

Similarly, the optimal acceleration of the jammer is given by

$$\mathbf{a}^{*}_{J} = -\mathbf{v}_J - C\frac{\|\mathbf{v}_J\|^2}{m} + \frac{\nabla J_J(\mathbf{v}_J)}{m(\mu_2 - \nabla J_J(e_J))}. \tag{30}$$

Lemma 3 (Uniqueness of NE): For the trajectory design subgame, where the power strategies P_{send} and P_{jamm} are fixed, if the following conditions are satisfied.

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1) Acceleration strategy sets $\{\mathbf{a}_{\rm BS} : \mathbf{a}_{\rm BS,min} \leq \mathbf{a}_{\rm BS} \leq \mathbf{a}_{\rm BS,max}\}$ and $\{\mathbf{a}_J : \mathbf{a}_{J,\min} \leq \mathbf{a}_J \leq \mathbf{a}_{J,\max}\}$ are compact and convex.
2) Utility functions $u_{\rm BS}$ and u_J are continuously differentiable with respect to accelerations in their domain.
3) At the NE point $(\mathbf{a}^*_{\rm BS}, \mathbf{a}^*_J)$, first-order optimality conditions hold.
4) The dominant diagonal condition is satisfied.

Then, the NE $(\mathbf{a}^*_{\rm BS}, \mathbf{a}^*_J)$ is locally unique in its neighborhood.

Proof: Please refer to Appendix B for detailed proof.

B. Lower Layer: Stackelberg Game of Power

After the acceleration strategies of the base station and the jammer are given, the optimal power strategies can be solved. Jointly considering the limited energy aboard and the acceleration strategies obtained from the upper layer game, the base station need to make the optimal decisions on the sending power $P_{\rm send}(t)$, and the jammer need to decide the optimal jamming power $P_{\rm jamm}(t)$. Higher jammer's jamming power would increase the success rate of jamming, but would also increase energy consumption and have an impact on the sending power strategy of the base station. In this subgame, we consider that both the base station and the jammer are smart, and either of them can intelligently respond to the other one's strategy.

Denote the power subgame as $G_l = \{B, J, P_{\rm send}, P_{\rm jamm}, u_{\rm BS}, u_J\}$, and we assume that the acceleration strategies of the players are fixed in this game. Since the jamming power of the jammer is usually set to a larger order of magnitude than the sending power of the base station, i.e., jamming power is tens of watts while sending power is a few watts or milliwatts, which means the states of the two players are different in the game, and the jammer plays a dominant role. It's reasonable to model this game as a Stackelberg game and set the jammer as the leader and the base station as the follower escaping from the jamming.

Specifically, the base station, the follower in the game, senses the strategy of the jammer and finds an optimal strategy to stay away from strong interference caused by the jammer. After that, the jammer senses the sending power of the base station and decides its jamming power. Till now, the jammer and the base station have finished a round of confrontation. As time progresses, the confrontation continues, and the game will eventually reach the open-loop Stackelberg equilibrium.

Definition 2 (Open-Loop Stackelberg Equilibrium): For the game $G_l = \{B, J, P_{\rm send}, P_{\rm jamm}, u_{\rm BS}, u_J\}$, strategy combination $\{P^*_{\rm send}, P^*_{\rm jamm}\}$ is the Open-Loop Stackelberg equilibrium if and only if

$$\begin{aligned} &u_{\rm BS}\left(P^*_{\rm send}, P^*_{\rm jamm}\right) \geq u_{\rm BS}\left(P_{\rm send}, P^*_{\rm jamm}\right) \\ &u_J\left(P^*_{\rm send}, P^*_{\rm jamm}\right) \geq u_J\left(P^*_{\rm send}, P_{\rm jamm}\right) \end{aligned} \tag{31}$$

where $P^*_{\rm send}$ is the optimal sending power strategy of the base station and $P^*_{\rm send}$ is the optimal jamming power strategy of the jammer.

1) Base Station

To solve the maximization utility problem of the base station, denote the vector of the costate variables as $\lambda(\mathbf{t}) = [\lambda_1(t), \lambda_2(t), \lambda_3(t)]$, and the Hamiltonian function (28) could be rewritten as follows:

$$\begin{aligned} H_{\rm BS} = &\, g_{\rm BS}\left(\mathbf{l}_{\rm BS}, \mathbf{v}_{\rm BS}, \mathbf{a}_{\rm BS}, P_{\rm send}, t\right) + \lambda_1 \cdot \mathbf{v}_{\rm BS} \\ &+ \lambda_2 \cdot \mathbf{a}_{\rm BS} + \lambda_3 \cdot \left(P_{\rm send} + P_{\rm move,BS}\right). \end{aligned} \tag{32}$$

According to Pontryagin's maximum principle, we can obtain the optimal sending power strategy of the base station from the following equations:

$$\begin{aligned} &\frac{\partial H_{\rm BS}}{\partial P_{\rm send}} = 0 \\ &\dot{\lambda}_1 = -\frac{\partial H_{\rm BS}}{\partial \mathbf{l}_{\rm BS}} \\ &\dot{\lambda}_2 = -\frac{\partial H_{\rm BS}}{\partial \mathbf{v}_{\rm BS}} \\ &\dot{\lambda}_3 = -\frac{\partial H_{\rm BS}}{\partial e_{\rm BS}}. \end{aligned} \tag{33}$$

Then, the optimal sending power strategy can be expressed as

$$P^*_{\rm send}(t) = \frac{B}{\ln 2\,(\eta_1 - \lambda_3)} - \frac{I^n_{\rm jamm} + N}{GL^n_{\rm BS}}. \tag{34}$$

The Hamiltonian function (32) is concave with respect to $P_{\rm send}$, so the optimal sending power strategy is the unique solution.

2) Jammer

Define the Hamiltonian function of the jammer as follows:

$$\begin{aligned} H_J = &\, g_J\left(\mathbf{l}_J, \mathbf{v}_J, \mathbf{a}_J, P_{\rm jamm}, t\right) + \varphi_1 \cdot \mathbf{v}_J + \varphi_2 \cdot \mathbf{a}_J \\ &+ \varphi_3 \cdot \left(P_{\rm jamm} + P_{{\rm move},J}\right) + \sum_{i=1}^{3} \pi_i \dot{\lambda}_i \end{aligned} \tag{35}$$

where $\pi(t) = [\pi_1(t), \pi_2(t), \pi_3(t)]$ is the costate variable of the costate function $\dot{\lambda}(t)$, and this is because the jammer plays the role of the leader in the game.

Then, similar to the base station, we can solve for the optimal jamming power strategy using the following equations:

$$\begin{aligned} &\frac{\partial H_J}{\partial P_{\rm jamm}} = 0 \\ &\dot{\varphi}_1 = -\frac{\partial H_J}{\partial \mathbf{l_J}} \\ &\dot{\varphi}_2 = -\frac{\partial H_J}{\partial \mathbf{v}_J} \\ &\dot{\varphi}_3 = -\frac{\partial H_J}{\partial e_J} \\ &\dot{\pi}_i = -\frac{\partial H_J}{\partial \lambda_i}, \quad i \in \{1, 2, 3\}. \end{aligned} \tag{36}$$

The optimal jamming power strategy for the jammer is

$$P^*_{\rm jamm}(t) = \frac{-\mathcal{A} + \sqrt{\Delta}}{2\delta g^n_{\rm jamm}} - \frac{N}{\delta g^n_{\rm jamm}} \tag{37}$$

where $\mathcal{A} = P_{\rm send}(t) GL^n_{\rm BS}$ and $\Delta = \mathcal{A}^2 + (4\delta g^n_{\rm jamm} B\mathcal{A}/ln2(\mu_1 - \varphi_3))$.

Remark: The lower layer power $P^*_{\rm send}(t)$ and the $P^*_{\rm jamm}(t)$ derived below depend on the leader's trajectory $[\mathbf{x}(t) = \mathbf{l}_{\rm BS}(t), \mathbf{v}_{\rm BS}(t)]$ through channel gain and energy related terms.

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These terms are continuous with respect to position/velocity and remain bounded under the physical constraints on states and controls. Consequently, the resulting power equilibria change continuously as the trajectory evolves within the feasible range, without abrupt jumps or instability. This clarifies the sensitivity of the lower layer equilibrium to the upper layer trajectory and keeps the hierarchical coupling well-behaved. A formal stability analysis of the adaptive update will be provided later in Section V.

V. ADP-Based Method

We can obtain the acceleration strategies of the base station and the jammer according to (29) and (30), but it is difficult to obtain the value function J_* by solving the HJB equations directly because of their high computational complexity. It is widely recognized that ADP provides an effective framework to approximate the value function and thus avoid the computational burden of solving the HJB equation analytically. Therefore, in this article, we adopt an ADP-based approach to approximate J_*. We focus on the analysis of the base station's acceleration strategy, while the analysis for other strategies follows a similar process.

ADP is a learning-based dynamic optimization framework that approximates the solution of the HJB equation through an iterative policy evaluation and policy improvement process. In this article, an actor–critic ADP structure is employed. The critic network estimates the state value function of the base station, and the acceleration strategy is iteratively improved based on the updated value function until convergence.

Through the critic neural network, the value function of the base station $J_{\rm BS}$ can be rewritten as

$$J_{\rm BS} = W_c^T \sigma_c + \varepsilon \tag{38}$$

where $W_c \in \mathbb{R}_l$ is the ideal weight, l is the number of neurons in the hidden layer, σ_c is the activation function of the critic network, and ε is the approximation error.

Then, the partial derivative of $J_{\rm BS}$ w.r.t $\mathbf{v}_{\rm BS}$ can be written as

$$\nabla J_{\rm BS}(\mathbf{v}_{\rm BS}) = \nabla\sigma_c(\mathbf{v}_{\rm BS})^T W_c + \nabla\varepsilon(\mathbf{v}_{\rm BS}) \tag{39}$$

and the partial derivate of $J_{\rm BS}$ with respect to $e_{\rm BS}$ can be written as

$$\nabla J_{\rm BS}(e_{\rm BS}) = \nabla\sigma_c(e_{\rm BS})^T W_c + \nabla\varepsilon(e_{\rm BS}). \tag{40}$$

Based on (39) and (40), the optimal acceleration strategy of the base station is

$$\begin{aligned}\mathbf{a}_{\rm BS}^* = &-\mathbf{v}_{\rm BS} - C\frac{\|\mathbf{v}_{\rm BS}\|^2}{m}\\ &+ \frac{\left(\nabla\sigma_c(\mathbf{v}_{\rm BS})^T W_c + \nabla\varepsilon(\mathbf{v}_{\rm BS})\right)}{m\left[\eta_2 - \left(\nabla\sigma_c(e_{\rm BS})^T W_c + \nabla\varepsilon(e_{\rm BS})\right)\right]}.\end{aligned} \tag{41}$$

By substituting the above equations into the Hamiltonian function $H_{\rm BS}$, we can rewrite the function.

But the ideal weight W_c is unknown, so we replace it with the approximate weight, which is denoted as $\hat{W}_c$. And the approximate value function $\hat{J}_{\rm BS}$ can be expressed as

$$\hat{J}_{\rm BS} = \hat{W}_c^T \sigma_c. \tag{42}$$

Algorithm 1 Optimal Acceleration Strategy Solving Algorithm for Base Station Based on ADP

Initialization: Let the iteration index $i = 0$, set the initial policy as $\mathbf{a}_{\rm BS}^0$, the initial weight of the critic network as $\hat{W}_c^i$, the computation precision as $\xi > 0$

1. $i = i + 1$, solve $\hat{J}_{\rm BS}^i$ according to the equation $\hat{H}_{\rm BS} = 0$
2. Update the acceleration strategy $\mathbf{a}_{\rm BS}^{\mathbf{i}}$ by (43)
3. If $|\hat{J}_{\rm BS}^i - \hat{J}_{\rm BS}^{i-1}| < \xi$, stop and set the strategy $\mathbf{a}_{\rm BS}^{\mathbf{i}}$ as optimal acceleration strategy; otherwise, go back to Step1.

Then, the optimal acceleration strategy of the base station can be approximated as

$$\hat{\mathbf{a}}_{\rm BS}^* = -\mathbf{v}_{\rm BS} - C\frac{\|\mathbf{v}_{\rm BS}\|^2}{m} + \frac{\nabla\sigma_c(\mathbf{v}_{\rm BS})^T \hat{W}_c}{m\left[\eta_2 - \nabla\sigma_c(e_{\rm BS})^T \hat{W}_c\right]}. \tag{43}$$

Based on the approximated value function and the optimal acceleration strategy, we can get the approximated HJB function $\hat{H}_{\rm BS}$

$$\begin{aligned}\hat{H}_{\rm BS} = &\, g_{\rm BS}(\mathbf{l}_{\rm BS}, \mathbf{v}_{\rm BS}, \mathbf{a}\hat{}_{\rm BS}, P_{\rm send}, t)\\ &+ \nabla\sigma_c(\mathbf{l}_{\rm BS})^T \hat{W}_c \cdot \mathbf{v}_{\rm BS} + \nabla\sigma_c(\mathbf{v}_{\rm BS})^T \hat{W}_c \cdot \hat{\mathbf{a}}_{\rm BS}^*\\ &+ \nabla\sigma_c(e_{\rm BS})^T \hat{W}_c \cdot \left(P_{\rm send} + \hat{P}_{\rm move,BS}\right).\end{aligned} \tag{44}$$

Define the error between the approximated and the true Hamiltonians as $e_c \triangleq \hat{H}_{\rm BS} - H_{\rm BS}$ and the objective optimal function of the critic network as

$$E\left(\hat{W}_c\right) = \frac{1}{2}e_c e_c^T. \tag{45}$$

Inspired by [45], we design the update law of the weight $\hat{W}_c$ as

$$\begin{aligned}\dot{\hat{W}}_c &= -\phi\frac{1}{\left(1+\omega^T\omega\right)^2}\frac{\partial E}{\partial \hat{W}_c}\\ &= -\phi\frac{1}{\left(1+\omega^T\omega\right)^2}\frac{\partial E}{\partial e_c}\frac{\partial e_c}{\partial \hat{W}_c}\\ &= -\phi\frac{\omega}{\left(1+\omega^T\omega\right)^2}e_c\end{aligned} \tag{46}$$

where ϕ denotes the learning rate of the critic network and $\omega = \nabla\sigma_c(\mathbf{l}_{\rm BS})\mathbf{v}_{\rm BS} + \nabla\sigma_c(\mathbf{v}_{\rm BS})\hat{\mathbf{a}}_{\rm BS} + \nabla\sigma_c(e_{\rm BS})(P_{\rm send} + \hat{P}_{\rm move,BS})$.

Then, the optimal acceleration strategy solving process of the base station can be summarized as Algorithm 1. The algorithm follows the policy iteration structure of ADP: it first performs policy evaluation (Step 1), where the critic network updates its weights $\hat{W}_c$ to approximate the value function. Then, it performs policy improvement (Step 2), where the actor's acceleration strategy $\mathbf{a}_{\rm BS}$ is updated using the new critic weights. This process repeats until the strategy converges.

Define the weight estimation error of the critic neural network as $\tilde{W}_c = W_c - \hat{W}_c$. Then, the weight estimation error dynamics are [46]

$$\begin{aligned}\dot{\tilde{W}}_c &= -\dot{\hat{W}}_c\\ &= -\phi\frac{\omega}{\left(1+\omega^T\omega\right)^2}\left(\omega^T\tilde{W}_c - \tau\right)\end{aligned} \tag{47}$$

where $\tau = -\nabla\varepsilon(\mathbf{l}_{\rm BS})^T\mathbf{v}_{\rm BS} - \nabla\varepsilon(\mathbf{v}_{\rm BS})^T\hat{\mathbf{a}}_{\rm BS} - \nabla\varepsilon(e_{\rm BS})(P_{\rm send} + \hat{P}_{\rm move,BS})$ is the residual error.

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Let the state-related vector be defined as

$$\chi = \begin{bmatrix} \mathbf{v}_{\text{BS}} \\ \hat{\mathbf{a}}_{\text{BS}} \\ P_{\text{send}} + \hat{P}_{\text{move,BS}} \end{bmatrix} \tag{48}$$

and $\tau = -\nabla\varepsilon \cdot \chi$.

According to (39) and (40), the gradient of the true value function J_{BS} can be expressed as

$$\nabla J_{\text{BS}} = (\nabla\sigma_c)^T W_c + \nabla\varepsilon \tag{49}$$

while the gradient of the approximated value function $\hat{J}_{\text{BS}}$, estimated by the critic network, is

$$\nabla \hat{J}_{\text{BS}} = (\nabla\sigma_c)^T \hat{W}_c. \tag{50}$$

Hence, the difference between the actual and approximated Hamilton functions can be written as

$$\begin{aligned} H_{\text{BS}} - \hat{H}_{\text{BS}} &= \left(\nabla J_{\text{BS}} - \nabla \hat{J}_{\text{BS}}\right) \cdot \chi \\ &= \left((\nabla\sigma_c)^T \left(W_c - \hat{W}_c\right) + \nabla\varepsilon\right) \cdot \chi \\ &= \omega^T \tilde{W}_c - \tau \end{aligned} \tag{51}$$

where $\omega = (\nabla\sigma_c)^T \chi$.

Then, the critic error defined in (44) can be further expressed as

$$e_c = \hat{H}_{\text{BS}} - H_{\text{BS}} = \tau - \omega^T \tilde{W}_c. \tag{52}$$

Taking the time derivative yields

$$\dot{e}_c = \dot{\tau} - \dot{\omega}^T \tilde{W}_c - \omega^T \dot{\tilde{W}}_c. \tag{53}$$

And from (47) we have

$$\dot{\tilde{W}}_c = \frac{\phi\, \omega}{\left(1 + \omega^T \omega\right)^2} e_c. \tag{54}$$

Assumption 1: To facilitate the stability analysis, we introduce the following basic assumptions that are commonly satisfied in neural network-based control systems [47], [48], [49].

1) The gradients of the activation function in the critic network are norm-bounded, i.e., $||\nabla\sigma_c(\mathbf{l}_{\text{BS}})|| < \delta_{c,Ml}$, $||\nabla\sigma_c(\mathbf{v}_{\text{BS}})|| < \delta_{c,Mv}$, and $||\nabla\sigma_c(e_{\text{BS}})|| < \delta_{c,Me}$.

2) The approximation error of the critic network and its gradient are norm-bounded, i.e., $||\varepsilon|| < \epsilon_\varepsilon$ and $||\nabla\varepsilon|| < \epsilon_{\nabla\varepsilon}$.

3) The initial weight of the critic network satisfies $||\hat{W}_c(0)|| \le \omega_{c,0}$, where $\omega_{c,0}$ is a positive constant.

Remark 1: Assumption 1 (3) only constrains the initial condition. The uniform boundedness of $\hat{W}_c(t)$ for all $t \ge 0$ will be established in Theorem 1.

Theorem 1: Under Assumption 1, when the weight of the critic network $\hat{W}_c$ is updated according to (46), the following properties hold.

1) The weight estimation error $\tilde{W}_c$ and the HJB approximation error e_c are uniformly ultimately bounded (UUB) and converge to a residual set.

2) The estimated critic network weight $\hat{W}_c(t)$ remains bounded for all $t \ge 0$.

Proof: Define the Lyapunov function candidate as

$$V = \frac{1}{2}\tilde{W}_c^\top \Gamma^{-1} \tilde{W}_c + \frac{\kappa}{2} e_c^2 \tag{55}$$

where $\Gamma = \gamma^{-1} I$ with $\gamma > 0$ and $\kappa > 0$. Then, the time derivative of V can be written as

$$\begin{aligned} \dot{V} &= \tilde{W}_c^\top \Gamma^{-1} \dot{\tilde{W}}_c + \kappa e_c \dot{e}_c \\ &= \frac{\phi}{(1 + \omega^\top \omega)^2} \tilde{W}_c^\top \Gamma^{-1} \omega e_c + \kappa e_c \dot{\tau} \\ &\quad - \kappa e_c \dot{\omega}^\top \tilde{W}_c - \kappa \frac{\phi ||\omega||^2}{(1 + \omega^\top \omega)^2} e_c^2. \end{aligned} \tag{56}$$

Applying Cauchy–Young inequalities to the three mixed terms (for arbitrary $\eta_1, \delta_1, \delta_2 > 0$) and substituting into (56) yield

$$\begin{aligned} \dot{V} \le &\left[\frac{\phi\gamma\eta_1}{2\,(1 + \omega^\top \omega)^2} ||\omega||^2 + \frac{\kappa}{2\delta_2} ||\dot{\omega}||^2\right] ||\tilde{W}_c||^2 \\ &- \left[\kappa \frac{\phi ||\omega||^2}{(1 + \omega^\top \omega)^2} - \frac{\phi\gamma}{2\eta_1\,(1 + \omega^\top \omega)^2} - \frac{\kappa\,(\delta_1 + \delta_2)}{2}\right] e_c^2 \\ &+ \frac{\kappa}{2\delta_1} ||\dot{\tau}||^2. \end{aligned} \tag{57}$$

Define

$$c_W(t) = \frac{\phi\gamma\eta_1}{2\,(1 + \omega^\top \omega)^2} ||\omega||^2 + \frac{\kappa}{2\delta_2} ||\dot{\omega}||^2 \tag{58}$$

$$c_e(t) = \kappa \frac{\phi ||\omega||^2}{(1 + \omega^\top \omega)^2} - \frac{\phi\gamma}{2\eta_1\,(1 + \omega^\top \omega)^2} - \frac{\kappa\,(\delta_1 + \delta_2)}{2} \tag{59}$$

$$\beta_0(t) = \frac{\kappa}{2\delta_1} ||\dot{\tau}||^2 \tag{60}$$

so that

$$\dot{V} \le c_W(t)\, ||\tilde{W}_c||^2 \;-\; c_e(t)\, e_c^2 \;+\; \beta_0(t). \tag{61}$$

Under Assumption 1 and the boundedness of system trajectories established in Lemma 1, the variables ω, τ, and their derivatives are bounded. Hence, there exist constants $\bar{c}_W$ and $\bar{\beta}$ satisfying $c_W(t) \le \bar{c}_W$ and $\beta_0(t) \le \bar{\beta}$. Moreover, if there exist $\alpha_0 > 0$ and $T_0 > 0$ satisfying the windowed excitation condition

$$\int_t^{t+T_0} \frac{\omega(\tau)\,\omega(\tau)^\top}{\left(1 + ||\omega(\tau)||^2\right)^2} d\tau \ge \alpha_0 I \tag{62}$$

and if the parameters $\eta_1, \delta_1, \delta_2, \gamma, \kappa$ are chosen such that

$$\int_t^{t+T_0} c_e(\tau)\, d\tau \ge \underline{c}_e T_0 > 0 \tag{63}$$

then there exist positive constants $\alpha_1, \alpha_2, \beta$ satisfying

$$\dot{V} \le -\alpha_1 ||\tilde{W}_c||^2 - \alpha_2 e_c^2 + \beta. \tag{64}$$

By the comparison lemma, $V(t)$ is UUB, and thus $\tilde{W}_c$ and e_c are UUB.

Remark: Even under strong jamming attacks, the boundedness of the system trajectories and energy constraints ensure that the critic-network learning remains within a compact set. Thus, while performance may degrade, the learned strategy remains stable and well defined.

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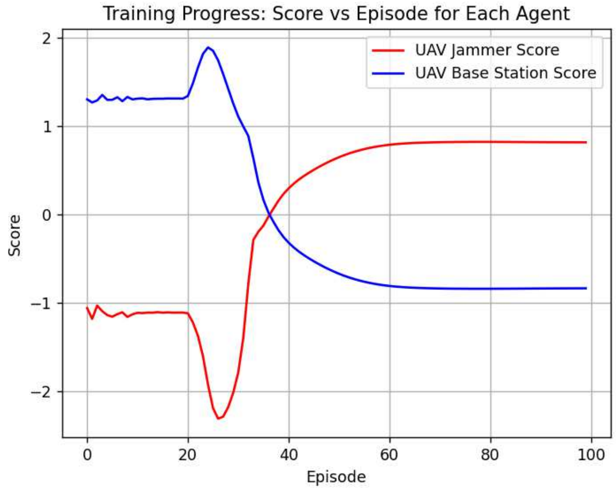


Fig. 3. Episode cumulative rewards of the UAV base station and the UAV jammer.

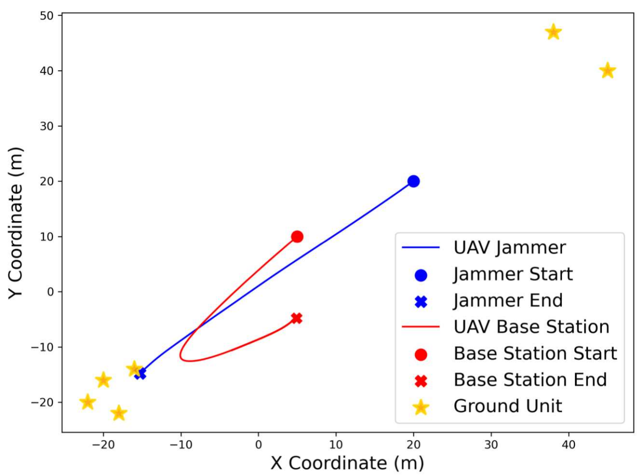
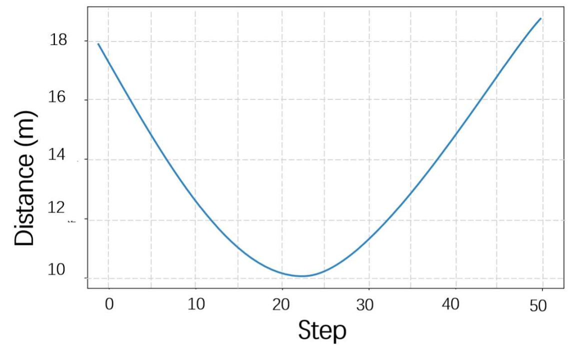


Fig. 4. Trajectories of the UAV base station and the UAV jammer.

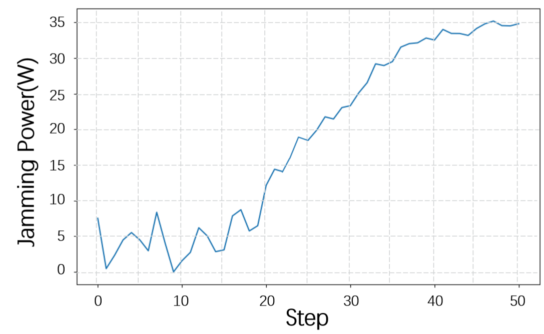
VI. Numerical Simulations

In this section, the effectiveness of the proposed model will be verified through relevant simulation experiments. The results of our simulations provide valuable insights into the optimization of resource allocation and decision-making processes in UAV communication networks.

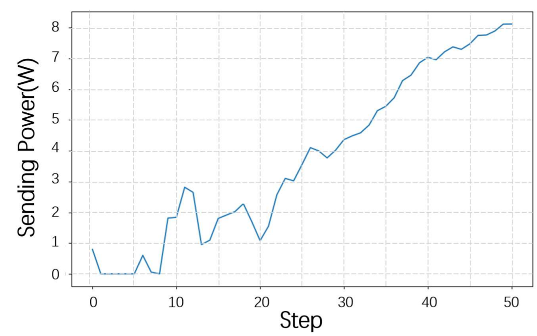
The simulation topology includes one UAV base station, one UAV jammer, and six ground units. They are distributed in an area of 100×100 m^2. The ground units were divided into two functional groups: a dense cluster with four nodes randomly distributed around the center $(-20, -20)$ and a sparse cluster with two nodes centered at (45, 45). Both the UAV base station and UAV jammer maintain a constant altitude of 10 m throughout the simulation. At the initial time $t = 0$, the position of the UAV base station is set to (5, 10), and the position of the UAV jammer is set to (20, 20). The initial velocity of the UAV base station is $[-10, -12]$ m/s. The initial velocity vector reflects its inherent strategy to prioritize connectivity with high-density clusters, where concentrated nodes enable reduced single-hop distance, consequently increasing channel



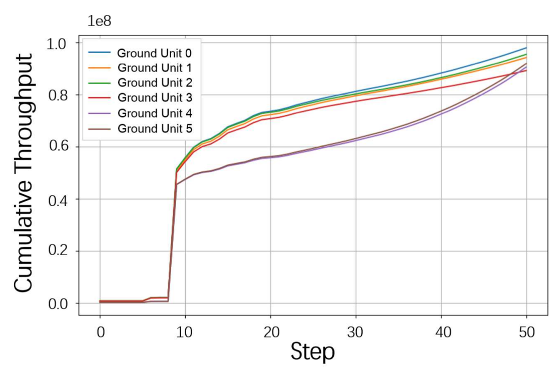
(a)



(b)



(c)



(d)

Fig. 5. Simulation results of the scenarios in Fig. 4. (a) Distance between the UAV base station and the UAV jammer. (b) Jamming power of the UAV jammer. (c) Sending the power of the UAV base station. (d) Cumulative throughputs of the ground units.

capacity. And the initial velocity of the UAV jammer is $[-12, -12]$ m/s, which can be represented as $\mathbf{v_J}(0) = (\mathbf{l_J}(0) - \mathbf{l}_{\mathrm{BS}}(0)/\|\mathbf{l_J}(0) - \mathbf{l}_{\mathrm{BS}}(0)\|)$, as the UAV jammer always attempts to

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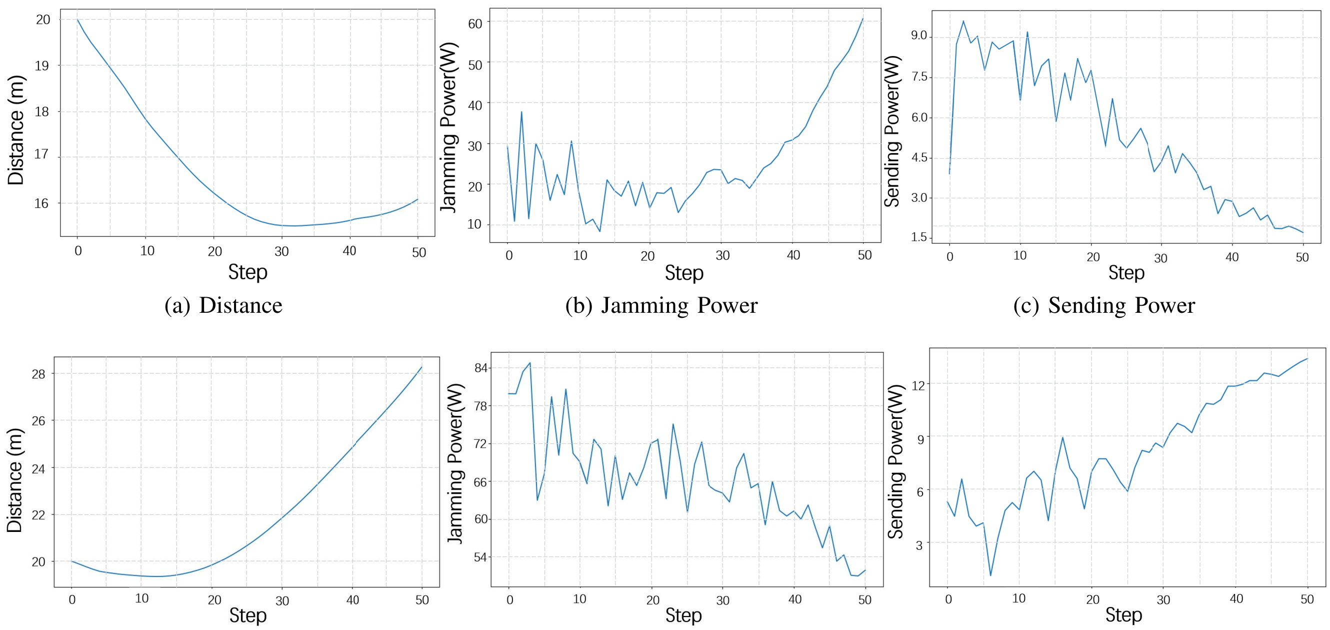

(a) Distance (b) Jamming Power (c) Sending Power

(d) Distance (e) Jamming Power (f) Sending Power

Fig. 6. Simulation results for different initial speeds. (a)–(c) Case I. (d)–(f) Case II.

track the UAV base station. The discretization time interval is set to 0.1 seconds. The mass of both the UAV base station and the UAV jammer is 5 kg. The initial energy of both is 10 000 units. The probability of LOS communication is 0.7, the path loss coefficient is $\alpha = 10^{-3}$, and the background noise power is −110 dBm/Hz.

The critic network is designed as a two-layer fully connected neural network (hidden layers of 128 and 64 neurons with ReLU activation and one linear output neuron. The parameters are trained using the Adam optimizer with a learning rate of 10^{-3} and a batch size of 256. The proposed framework performs off-policy actor–critic updates with experience replay at each time step. Each update requires a single forward–backward pass of moderately sized neural networks, resulting in a computational cost proportional to the number of trainable parameters, i.e., $\mathcal{O}(B \cdot n_w)$ where B is the batch size and n_w denotes the parameter count of the critic network. As the actor is evaluated in inference mode for decision making, its contribution to latency is negligible compared with the critic update. This update scheme keeps the computational overhead low and the decision latency well within the real-time control cycle of UAV processors, ensuring feasibility for onboard deployment. The simulation was conducted using the aforementioned parameters over 100 training episodes, with each episode comprising 50 time steps. The evolution of episode cumulative rewards for both the UAV base station and UAV jammer throughout the training process is illustrated in Fig. 3. The reward exhibits an inverse relationship between the UAV base station and UAV jammer, reflecting their inherent adversarial nature in this strategic game. Both agents demonstrate progressive convergence in their episode cumulative rewards as training advances, indicating that their policies gradually approach an equilibrium. This stabilization pattern suggests the emergence of mutually optimal response strategies through iterative learning.

Simulate the interaction process between the UAV base station and the UAV jammer using the obtained model. Fig. 4 illustrates the movement trajectories of the UAV base station and the UAV jammer over 50 time steps. As depicted in the figure, the UAV base station and the UAV jammer exhibit a typical pursuit-evasion relationship. Initially, the UAV base station moves toward the dense cluster, with the UAV jammer closely following. The UAV jammer reaches the vicinity of the dense cluster ahead of the UAV base station and decelerates to hover, thereby intercepting the UAV base station. Upon detecting the interception by the jammer, the UAV base station alters its course and heads toward the sparse cluster, attempting to serve the ground units with less interference.

In the scenario depicted in Fig. 4, the variation in distance between the UAV base station and the UAV jammer is shown in Fig. 5(a), while the jamming power of the UAV jammer is presented in Fig. 5(b). The sending power of the UAV base station is illustrated in Fig. 5(c), and the cumulative throughputs between the UAV base station and the ground units are depicted in Fig. 5(d). At the initial stage of the simulation, the UAV jammer effectively pursues the UAV base station, leading to a gradual decrease in their distance. Subsequently, the UAV jammer enters a hovering phase to block the UAV base station, prompting the UAV base station to alter its trajectory to evade, resulting in an increase in its distance once again.

Correspondingly, during the initial stage, the UAV jammer maintains a low jamming power level to conserve energy, as it is positioned far from the UAV base station and the base station operates at low sending power. Around step 15, as the distance between the UAV jammer and the UAV base station significantly decreases and the base station initiates data transmission, the UAV jammer gradually increases its jamming power to disrupt the UAV base station's communication. Around step 40, the UAV jammer gradually slows

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down the rate of jamming power increase. At this stage, as the distance between the UAV jammer and the UAV base station expands again, the effectiveness of further increasing the jamming power diminishes.

At the beginning of this process, the UAV base station is relatively distant from both clusters of ground units, resulting in a lower transmission power to conserve energy. As the simulation progresses, the UAV base station gradually moves closer to the densely distributed ground units and detects no significant interference. Consequently, around step 9, it begins to increase its sending power for data transmission. Around step 20, as the UAV jammer begins introducing interference, the UAV base station further raises its sending power to maintain communication quality with the ground units. Subsequently, as the jamming power continues to intensify, the UAV base station correspondingly increases its sending power to counteract the interference effects.

In the early phase, due to the low sending power of the UAV base station, the data throughput between the UAV base station and the ground units increases at a slow rate. Around step 9, as the UAV base station increases its sending power, the cumulative throughput of all ground units rises sharply. At this point, the UAV base station is closer to the densely distributed ground units, leading to a higher cumulative throughput for ground units 0–3 compared to units 4 and 5. In the latter half of the simulation, as the UAV base station changes its trajectory toward the sparsely distributed ground units, the growth rate of cumulative throughput for units 4 and 5 gradually surpasses that of units 0–3.

To investigate the power strategy interactions between the UAV base station and the UAV jammer, simulations were conducted for two different scenarios. The variations in the distance between the UAV base station and the UAV jammer, the jamming power of the UAV jammer, and the sending power of the UAV base station during the simulation are illustrated in Fig. 6. In Case I, the UAV jammer has a higher initial speed than the UAV base station, allowing it to effectively pursue the UAV base station, resulting in a gradual decrease in the distance between them. Since the UAV jammer and the UAV base station are initially positioned far apart, the UAV jammer operates at a relatively low jamming power. In response, the UAV base station rapidly increases its sending power to enhance data throughput. As the simulation progresses, the UAV jammer moves closer to the UAV base station, gradually reducing the distance between them, which reaches its minimum around the 30th step. During this pursuit phase, the jammer's power remains at a relatively low level with minor fluctuations. After reaching the minimum distance, the jammer begins to increase its jamming power, disrupting the communication of the UAV base station. Throughout the simulation, the UAV base station continuously decreases its sending power as the jammer approaches and its jamming power intensifies, aiming to reduce energy consumption while maintaining operational efficiency. In Case II, the initial velocity of the UAV jammer is reduced to create favorable conditions for the UAV base station to evade pursuit. As the simulation progresses, the distance between the two UAVs gradually increases. To conserve energy, the UAV jammer systematically decreases its jamming power, while the UAV base station correspondingly amplifies its transmission power to strengthen communication with ground units. The above results indicate that during the dynamic interaction between the UAV base station and the UAV jammer, the base station can adaptively adjust its communication strategy based on the jammer's position and interference power, thereby optimizing communication efficiency.

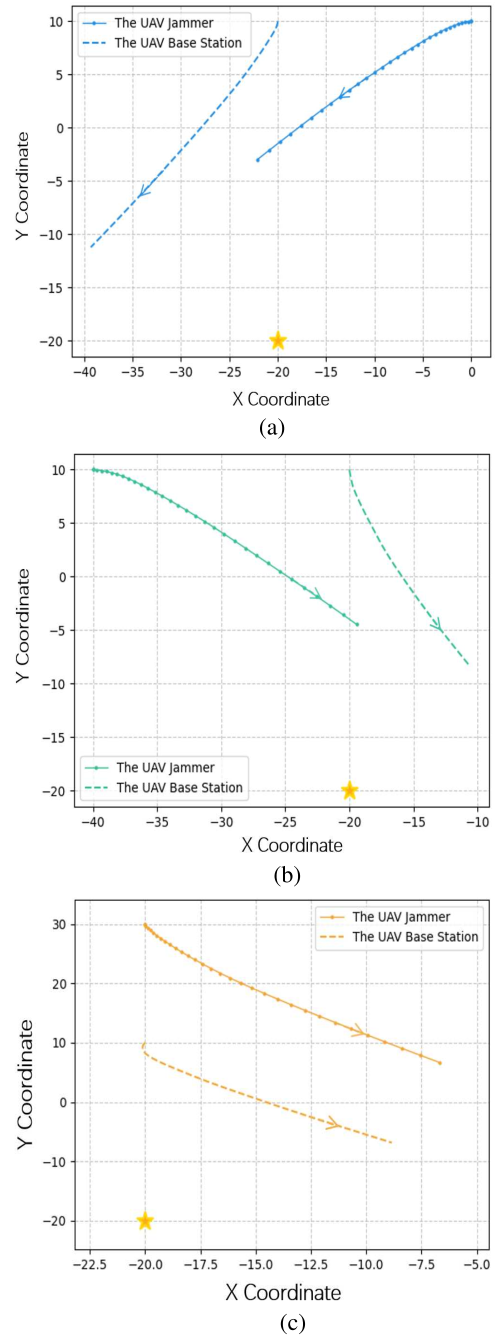

Fig. 7. Trajectories of the UAV base station and jammer for different initial relative positions. (a) Case I $\mathbf{l_J}(0) = [0, 10]$. (b) Case II $\mathbf{l_J}(0) = [-40, 10]$. (c) Case III $\mathbf{l_J}(0) = [0, 30]$.

Then, we conducted tests on the trajectories of UAV base stations and UAV jammers under different initial relative positions, as illustrated in Fig. 7. For all three scenarios, the

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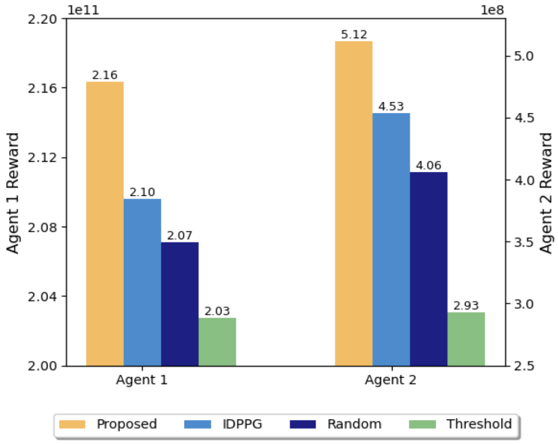

Fig. 8. Cumulative rewards for the UAV base station and the UAV jammer.

UAV base station was initially positioned at [−20, 10], the only ground unit at [−20, −20], while the UAV jammer occupied three distinct starting locations: [0, 10], [−40, 10], and [0, 30]. Observations from the figure reveal that the UAV base station successfully maneuvered toward the ground unit in all cases while evading pursuit from the UAV jammer. Concurrently, the UAV jammer demonstrated effective tracking and interception capabilities against the UAV base station.

We compared the proposed method with several baseline strategies: independent deep deterministic policy gradient (IDDPG), random, and threshold-based strategies. Fig. 8 shows the cumulative rewards for the UAV jammer and the UAV base station over 50 time steps under these strategies. In the IDPPG strategy, each UAV independently uses a DDPG agent to learn its policy without explicit knowledge or coordination with the other agents' policies. In the random strategy, UAVs randomly choose actions without considering the state-action relationship. The threshold strategy decides based on the distance between the UAVs: the UAV base station's transmission power is set to 0 when the distance is below α m, and the UAV jammer's interference power is set to 0 when the distance exceeds β m. This is because when the UAVs are close, the UAV jammer's interference can significantly reduce the ground unit's SNR, making signal interpretation difficult, so the UAV base station pauses transmission to save energy. Conversely, when they are far apart, the jamming effect is poor, and the UAV jammer turns off to save energy. As Fig. 8 shows, the proposed method outperforms the other strategies. Specifically, it increases the UAV jammer's reward by about 2.86% over IDPPG, 4.35% over the random strategy, and 6.40% over the threshold strategy. For the UAV base station, the proposed method achieves gains of about 13.02% over IDPPG, 26.11% over the random strategy, and 74.74% over the threshold strategy. These improvements are mainly due to the method's ability to dynamically adjust its decisions according to environmental changes and the opponent's behavior, rather than relying on fixed rules. Although it has a slightly higher computational complexity, the gains in communication efficiency and energy savings outweigh the additional computational cost, offering long-term comprehensive benefits.

VII. Conclusion

We propose a comprehensive system model for UAV adversarial communication scenarios, integrating communication, motion, and energy models. This model captures the interactions between UAV base stations and malicious drone jammers, considering factors such as interference and energy consumption. To address the complexity of the joint optimization problem, we develop a hierarchical game-theoretic framework that decomposes the problem into two subgames: a trajectory subgame and a power subgame. The trajectory subgame is formulated as a differential game to derive optimal acceleration strategies, while the power subgame is modeled as a Stackelberg game to determine optimal power allocation.

Simulation results demonstrate that the proposed approach effectively optimizes resource allocation and decision-making processes in UAV communication networks. The UAV base station and the UAV jammer exhibit typical pursuit-evasion behaviors, where the base station successfully evades jamming interference, while the UAV jammer effectively tracks and intercepts the base station. Compared with the IDDPG, random, and threshold-based strategies, the proposed method achieves significantly higher cumulative rewards, highlighting its superiority in dynamic decision-making and its enhanced adaptability to environmental and opponent states.

However, to establish a clear foundational analysis, this study adopts several simplifying assumptions that define the scope of our investigation. The framework is intentionally focused on a single base station-jammer pair, which is essential for isolating key variables but does not yet validate scalability in large-scale networks. Furthermore, the scenario assumptions are constrained: we consider a simplified channel model akin to single-channel jamming, UAVs operating at a fixed altitude rather than engaging in full 3-D maneuvering, and static ground units, which do not account for mobile targets (e.g., vehicular units). While these assumptions allow for a rigorous analysis of the core game-theoretic interactions, they limit the model's direct applicability to more complex, dynamic, and multidimensional real-world scenarios.

Future research will build upon this foundational work in a phased manner. The immediate next step is to extend our framework to multiagent scenarios, beginning with a small-scale multi-UAV cooperative communication system under a distributed jamming threat. This will allow us to validate the scalability of our core game-theoretic approach and investigate the emergent challenges of decentralized decision-making. As a subsequent phase of research, we plan to enhance the model's physical fidelity by incorporating full 3-D maneuvering and the dynamics of mobile ground units. Further avenues could then explore more complex adversarial strategies, such as multichannel parallel jamming. This staged approach ensures that each extension is built upon a methodologically sound and well-understood foundation.

Appendix A
Proof of Lemma 2

Define a value function as follows:

$$J_{\mathrm{BS}}\left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0\right)$$

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$$= \max_{\mathbf{a}_{\mathrm{BS}}} \left[\int_{t=t_0}^{T} g_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, \mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}, t \right) dt \right]. \tag{65}$$

Equation (65) can be converted into

$$\begin{aligned} &J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0 \right) \\ &= \max_{\mathbf{a}_{\mathrm{BS}}} \left[\int_{t=t_0}^{t_0+\Delta t} g_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, \mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}, t \right) dt \right. \\ &\qquad \left. + \int_{t=t_0+\Delta t}^{T} g_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, \mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}, t \right) dt \right] \end{aligned} \tag{66}$$

which is equal to

$$\begin{aligned} &J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0 \right) \\ &= \max_{\mathbf{a}_{\mathrm{BS}}} \left[\int_{t=t_0}^{t_0+\Delta t} g_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, \mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}, t \right) dt + J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0} \right.\right. \\ &\qquad \left.\left. + \Delta \mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}_0} + \Delta \mathbf{v}_{\mathrm{BS}}, e_{\mathrm{BS}_0} + \Delta e_{\mathrm{BS}}, t_0 + \Delta t \right) \right]. \end{aligned} \tag{67}$$

With the mean value theorems of definite integrals and Taylor's expansion, (67) can be written as

$$\begin{aligned} &J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0 \right) \\ &= \max_{\mathbf{a}_{\mathrm{BS}}} \left[g_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, \mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}, t \right) \Delta t \right. \\ &\quad + J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0 \right) \\ &\quad + \frac{\partial J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0 \right)}{\partial t} \Delta t \\ &\quad + \frac{\partial J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0 \right)}{\partial \mathbf{l}_{\mathrm{BS}}} \Delta \mathbf{l}_{\mathrm{BS}} \\ &\quad + \frac{\partial J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0 \right)}{\partial \mathbf{v}_{\mathrm{BS}}} \Delta \mathbf{v}_{\mathrm{BS}} \\ &\quad + \frac{\partial J_{\mathrm{BS}} \left(\mathbf{l}_{\mathrm{BS}_0}, \mathbf{v}_{\mathrm{BS}_0}, e_{\mathrm{BS}_0}, t_0 \right)}{\partial e_{\mathrm{BS}}} \\ &\quad \left. \cdot \left(P_{\mathrm{send}}(t) + P_{\mathrm{move,BS}}(t) \right) + O(\mathrm{h}) \right] \end{aligned} \tag{68}$$

where $O(h)$ denotes the higher order terms and can be omitted. Then, we have

$$\begin{aligned} 0 = \max_{\mathbf{a}_{\mathrm{BS}}} &[g_{\mathrm{BS}}(\mathbf{l}_{\mathrm{BS}}, \mathbf{v}_{\mathrm{BS}}, \mathbf{a}_{\mathrm{BS}}, P_{\mathrm{send}}, t)\Delta t + \nabla J_{\mathrm{BS}}(t) \cdot \Delta t \\ &+ \nabla J_{\mathrm{BS}}(\mathbf{l}_{\mathrm{BS}}) \cdot \Delta \mathbf{l}_{\mathrm{BS}} + \nabla J_{\mathrm{BS}}(\mathbf{v}_{\mathrm{BS}}) \cdot \Delta \mathbf{v}_{\mathrm{BS}} \\ &+ \nabla J_{\mathrm{BS}}(e_{\mathrm{BS}}) \cdot (P_{\mathrm{send}}(t) + P_{\mathrm{move,BS}}(t)). \end{aligned} \tag{69}$$

Dividing the resulting equation by Δt with Δt approaching 0, we yield (26), and the proof process of the UAV-jammer is similar to that of the base station.

Then, the proof is finished. □

APPENDIX B
PROOF OF LEMMA 3

At the NE, each player optimizes its own utility function. If the equilibrium point is in the interior of the strategy set, the first-order conditions must be satisfied: For the base station

$$\left.\frac{\partial u_{\mathrm{BS}}}{\partial \mathbf{a}_{\mathrm{BS}}}\right| \left(\mathbf{a}_{\mathrm{BS}}^*, \mathbf{a}_J^* \right) = \mathbf{0}. \tag{70}$$

For the jammer

$$\left.\frac{\partial u_J}{\partial \mathbf{a}_J}\right| \left(\mathbf{a}_{\mathrm{BS}}^*, \mathbf{a}_J^* \right) = \mathbf{0}. \tag{71}$$

We denote

$$F_{\mathrm{BS}} \left(\mathbf{a}_{\mathrm{BS}}^*, \mathbf{a}_J^* \right) = \mathbf{0} \tag{72}$$
$$F_J \left(\mathbf{a}_{\mathrm{BS}}^*, \mathbf{a}_J^* \right) = \mathbf{0} \tag{73}$$

where we define

$$F_{\mathrm{BS}} \left(\mathbf{a}_{\mathrm{BS}}, \mathbf{a}_J \right) := \frac{\partial u_{\mathrm{BS}}}{\partial \mathbf{a}_{\mathrm{BS}}} \tag{74}$$
$$F_J \left(\mathbf{a}_{\mathrm{BS}}, \mathbf{a}_J \right) := \frac{\partial u_J}{\partial \mathbf{a}_J}. \tag{75}$$

We now differentiate the first-order condition system. Construct the Jacobian matrix

$$\mathbf{J} = \begin{pmatrix} \frac{\partial F_{\mathrm{BS}}}{\partial \mathbf{a}_{\mathrm{BS}}} & \frac{\partial F_{\mathrm{BS}}}{\partial \mathbf{a}_J} \\ \frac{\partial F_J}{\partial \mathbf{a}_{\mathrm{BS}}} & \frac{\partial F_J}{\partial \mathbf{a}_J} \end{pmatrix}. \tag{76}$$

By the definition of second-order partial derivatives, this is equivalent to

$$J = \begin{pmatrix} H_{\mathrm{BS,BS}} & H_{\mathrm{BS},J} \\ H_{J,\mathrm{BS}} & H_{J,J} \end{pmatrix} \tag{77}$$

where

$H_{\mathrm{BS,BS}} = (\partial^2 u_{\mathrm{BS}} / \partial \mathbf{a}_{\mathrm{BS}}^2)$ is a 2×2 matrix;
$H_{\mathrm{BS},J} = (\partial^2 u_{\mathrm{BS}} / \partial \mathbf{a}_{\mathrm{BS}} \partial \mathbf{a}_J)$ is a 2×2 matrix;
$H_{J,\mathrm{BS}} = (\partial^2 u_J / \partial \mathbf{a}_J \partial \mathbf{a}_{\mathrm{BS}})$ is a 2×2 matrix;
$H_{J,J} = (\partial^2 u_J / \partial \mathbf{a}_J^2)$ is a 2×2 matrix.

The diagonal term $H_{\mathrm{BS,BS}}$ represents the second-order effect of the base station's acceleration decision on its own utility. The off-diagonal term $H_{\mathrm{BS},J}$ represents how the jammer's acceleration affects the base station's utility through communication quality. The Jacobian matrix satisfies the dominant diagonal condition, which means that the degree to which one's own decision affects oneself is greater than the influence of the opponent's decision. By symmetry, the same arguments apply to the jammer, ensuring both diagonal dominance conditions are satisfied.

For a block matrix $J = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, if A is invertible, then

$$\det(J) = \det(A) \cdot \det\left(D - CA^{-1}B \right). \tag{78}$$

In our case, we have $A = H_{\mathrm{BS,BS}}$, $B = H_{\mathrm{BS},J}$, $C = H_{J,\mathrm{BS}}$, $D = H_{J,J}$. Under the dominant diagonal condition, $\|H_{\mathrm{BS,BS}}\| > \|H_{\mathrm{BS},J}\|$ which implies that the eigenvalues of $H_{\mathrm{BS,BS}}$ dominate the cross terms to some extent. This guarantees $\det(H_{\mathrm{BS,BS}}) \neq 0$. Similarly, $\|H_{J,J}\| > \|H_{J,\mathrm{BS}}\|$ ensures $\det(H_{J,J}) \neq 0$. Thus, we can conclude that $\det(J) \neq 0$, meaning that the Jacobian matrix is nonsingular. (A rigorous proof requires matrix perturbation theory, but intuitively, the diagonal elements are strong enough to ensure the overall determinant is nonzero.)

Now consider the system of equations

$$F_{\mathrm{BS}} \left(\mathbf{a}_{\mathrm{BS}}, \mathbf{a}_J \right) = \mathbf{0} \tag{79}$$
$$F_J \left(\mathbf{a}_{\mathrm{BS}}, \mathbf{a}_J \right) = \mathbf{0}. \tag{80}$$

We have established that as follows.

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1) The point $(a_{\mathrm{BS}}^*, a_J^*)$ satisfies the system (i.e., it is a solution to the first-order conditions).
2) The Jacobian matrix at this point is nonsingular.

Since both conditions are satisfied, the implicit function theorem guarantees that in the neighborhood of $(a_{\mathrm{BS}}^*, a_J^*)$, the solution to the system of equations is unique. Since any NE must satisfy the first-order optimality condition, and the Implicit Function Theorem ensures that (a_{BS}, a_J) is the unique solution to these conditions in the neighborhood, we conclude that the NE is locally unique in this neighborhood.

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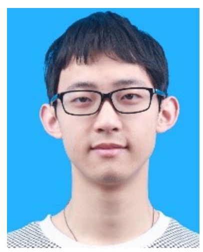

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